

Apex Racing

Formula 1 Team Database

CIS 245 - Database Design, Final Submission

Montgomery County Community College

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Section 1: Overview

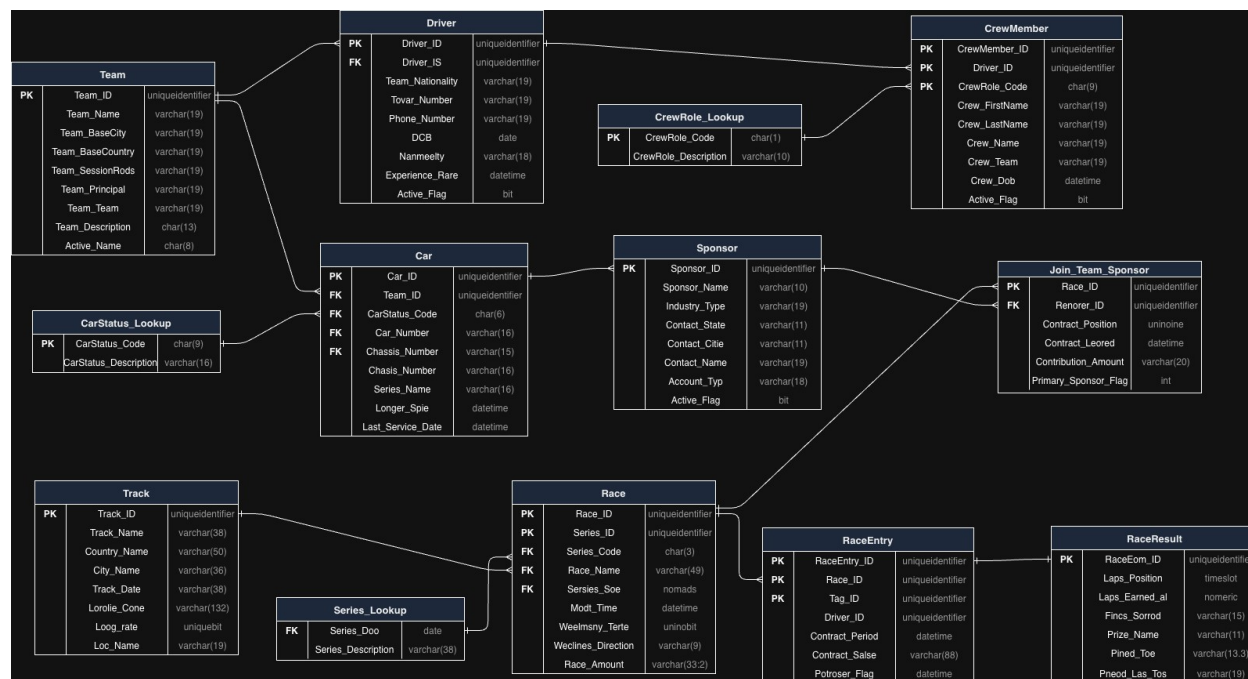
This is my full combined document for the Apex Racing F1 Team Database project covering all four CIS 245 labs. Lab 1 was the design phase, here I figured out the tables, relationships, and put together the E-R Diagram and Data Dictionary. Lab 2 was building the actual physical database with all the SQL. Lab 3 was writing 8 views against it. Lab 4 is the stored procedures phase. All SQL and documentation for each lab is included below in order.

Section 2: Lab 1 - Database Design

Lab 1 was the design phase for the Apex Racing database. I decided btw all the tables, figured how they relate to each other, and document everything before writing any code.

Section 2A: Physical IE Crow's Foot E-R Diagram

The Physical IE Crow's Foot E-R Diagram for the Apex Racing database was designed using draw.io. The diagram is included below. All 13 tables are shown with their fields, data types, mandatory indicators, primary keys, foreign keys, and notations.



Section 2B: Assumptions List

The following assumptions were made during the design of the E-R Diagram and the overall database schema:

- A Team may have many Drivers and many CrewMembers, but each Driver and each CrewMember belongs to exactly one Team at a time.
- A Driver may be entered into many Races over time through the RaceEntry table. Each RaceEntry links one Driver, one Car, and one Race.
- A Car belongs to exactly one Team. The same Car can appear in multiple RaceEntry records across different races.
- The RaceResult table has a one-to-one relationship with RaceEntry. Every RaceEntry that is completed should produce exactly one RaceResult row.
- The Join_Team_Sponsor table resolves the many-to-many relationship between Teams and Sponsors. A Team may have multiple Sponsors and a Sponsor may support multiple Teams.
- A Race is held at exactly one Track and belongs to exactly one racing Series. A Track may host many Races over multiple seasons.
- CrewRole_Lookup, CarStatus_Lookup, and Series_Lookup are reference tables used to enforce consistent coded values across the database.
- All primary keys are GUIDs (uniqueidentifier) generated by the DBMS to avoid sequential ID vulnerabilities and to support distributed data entry.
- A NULL Finish_Position in RaceResult indicates the driver did not finish. The Finish_Status field should always be populated to clarify the reason such as DNF, DNS, or DSQ.
- Team_PrincipalName is stored as a plain varchar in the Team table rather than as a foreign key to a person table, as team principals are not tracked with the same level of detail as drivers or crew members.
- Circular references are avoided. Team references no other main table through a mandatory foreign key. All relationships flow outward from Team to Driver, CrewMember, and Car.
- All datetime fields store values in UTC. Application layers are responsible for handling local timezone conversion.
- The Active_Flag bit field is used across Driver, CrewMember, Sponsor, and Track to allow soft deletion of records rather than physically removing data.

Section 2C: Data Dictionary

The data dictionary below defines each table in the Apex Racing database. It specifies all fields, data types, key designations, mandatory indicators, indexing, default values, and remarks. This data dictionary aligns directly with the Physical E-R Diagram presented in Section 2A.

Table 1: Team

Stores core identity and contact information for each racing team in the database.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
Team_ID	uniqueidentifier	PK	Y	Y	DBMS generated	Surrogate key (GUID)
Team_Name	varchar(60)		Y	Y		Official registered team name
Team_BaseCity	varchar(40)		Y			City where the team is headquartered
Team_BaseState	varchar(30)		Y			State or region where the team is based
Team_PrincipalName	varchar(60)		Y			Full name of the team principal
Team_Phone	varchar(20)					Primary team contact phone number
Team_Email	varchar(80)					Primary team contact email address
Team_FoundedYear	smallint		Y			4-digit year the team was founded
Team_Status	char(1)		Y		A	A=Active, I=Inactive

Table 2: Driver

Stores information about licensed racing drivers associated with a team.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
Driver_ID	uniqueidentifier	PK	Y	Y	DBMS generated	Surrogate key (GUID)
Team_ID	uniqueidentifier	FK	Y	Y		References Team
Driver_FirstName	varchar(30)		Y			Driver's legal first name
Driver_LastName	varchar(30)		Y	Y		Driver's legal last name
License_Number	varchar(25)		Y	Y		Unique racing license number
DOB	date		Y			Driver's date of birth

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
Nationality	varchar(40)		Y			Driver's nationality
Experience_Years	smallint				0	Years of professional racing experience
Active_Flag	bit		Y		1	1=Active, 0=Inactive

Table 3: CrewRole_Lookup

Lookup table defining valid crew role codes and their descriptions.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
CrewRole_Code	char(3)	PK	Y			3-character role code e.g. MEC, ENG, CHF
CrewRole_Description	varchar(40)		Y			Full description of the crew role

Table 4: CrewMember

Stores information about non-driver team personnel such as mechanics and engineers.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
CrewMember_ID	uniqueidentifier	PK	Y	Y	DBMS generated	Surrogate key (GUID)
Team_ID	uniqueidentifier	FK	Y	Y		References Team
CrewRole_Code	char(3)	FK	Y			References CrewRole_Lookup
Crew_FirstName	varchar(30)		Y			Crew member's legal first name
Crew_LastName	varchar(30)		Y	Y		Crew member's legal last name
Crew_Phone	varchar(20)					Crew member's contact phone number
Crew_Email	varchar(80)					Crew member's work email address
Hire_Date	datetime		Y			Date the crew member was hired
Active_Flag	bit		Y		1	1=Active, 0=Inactive

Table 5: CarStatus_Lookup

Lookup table defining valid car status codes and their descriptions.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
CarStatus_Code	char(2)	PK	Y			2-character status code e.g. AC, DC, SP, RP
CarStatus_Description	varchar(30)		Y			Full description of the car status

Table 6: Car

Stores details about each race car chassis assigned to a team.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
Car_ID	uniqueidentifier	PK	Y	Y	DBMS generated	Surrogate key (GUID)
Team_ID	uniqueidentifier	FK	Y	Y		References Team
CarStatus_Code	char(2)	FK	Y			References CarStatus_Lookup
Car_Number	varchar(10)		Y	Y		Racing number displayed on the car
Chassis_Number	varchar(30)		Y	Y		Manufacturer chassis serial number
Engine_Supplier	varchar(40)		Y			Name of the engine supplier
Model_Name	varchar(40)		Y			Car model or specification name
Model_Year	smallint		Y			Year of the car model
Last_Service_Date	datetime					Date the car last underwent scheduled service

Table 7: Sponsor

Stores information about corporate sponsors that may be associated with teams.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
Sponsor_ID	uniqueidentifier	PK	Y	Y	DBMS generated	Surrogate key (GUID)
Sponsor_Name	varchar(80)		Y	Y		Official sponsor company name

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
Industry_Type	varchar(40)		Y			Industry sector of the sponsor
Contact_Name	varchar(30)					Name of the primary sponsor contact
Contact_Email	varchar(80)					Email address of the sponsor contact
Contact_Phone	varchar(20)					Phone number of the sponsor contact
Sponsor_Tier	varchar(20)		Y			Title, Primary, or Associate
Active_Flag	bit		Y		1	1=Active, 0=Inactive

Table 8: Join_Team_Sponsor

Join table resolving the many-to-many relationship between Teams and Sponsors.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
Team_ID	uniqueidentifier	PK/FK	Y	Y		References Team
Sponsor_ID	uniqueidentifier	PK/FK	Y	Y		References Sponsor
Contract_StartDate	datetime		Y			Date the sponsorship contract began
Contract_EndDate	datetime		Y			Date the sponsorship contract ends
Contribution_Amount	numeric(12,2)		Y		0.00	Total financial contribution in USD
Primary_Sponsor_Flag	bit		Y		0	1=Primary sponsor, 0=Secondary

Table 9: Track

Stores information about racing tracks or circuits where events are held.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
Track_ID	uniqueidentifier	PK	Y	Y	DBMS generated	Surrogate key (GUID)
Track_Name	varchar(80)		Y	Y		Full official name of the track
Country_Name	varchar(40)		Y			Country where the track is located
City_Name	varchar(40)		Y			City nearest to the track
Track_Type	varchar(20)		Y			Road Course, Street Circuit, or Oval

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
Length_Miles	numeric(5,2)		Y			Length of one lap in miles
Turns_Count	smallint		Y			Number of turns per lap
FIA_Grade	varchar(10)					FIA safety grade designation
Active_Flag	bit		Y		1	1=Active venue, 0=No longer in use

Table 10: Series_Lookup

Lookup table defining valid racing series codes and their descriptions.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
Series_Code	char(4)	PK	Y			4-character series code e.g. F1, F2, FE
Series_Description	varchar(40)		Y			Full name of the racing series

Table 11: Race

Stores details about individual race events on the calendar.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
Race_ID	uniqueidentifier	PK	Y	Y	DBMS generated	Surrogate key (GUID)
Track_ID	uniqueidentifier	FK	Y	Y		References Track
Series_Code	char(4)	FK	Y	Y		References Series_Lookup
Race_Name	varchar(80)		Y			Official name of the race event
Season_Year	smallint		Y	Y		Championship season year
Race_Date	datetime		Y	Y		Scheduled date of the race
Scheduled_Laps	smallint		Y			Total number of laps scheduled
Weather_Summary	varchar(30)					Dry, Wet, or Mixed
Purse_Amount	numeric(12,2)				0.00	Total prize purse in USD

Table 12: RaceEntry

Stores each driver and car combination entered into a specific race.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
RaceEntry_ID	uniqueidentifier	PK	Y	Y	DBMS generated	Surrogate key (GUID)
Race_ID	uniqueidentifier	FK	Y	Y		References Race
Car_ID	uniqueidentifier	FK	Y	Y		References Car
Driver_ID	uniqueidentifier	FK	Y	Y		References Driver
Starting_Position	smallint		Y			Grid position at race start
Qualifying_Time	numeric(8,3)					Best qualifying lap time in seconds
Entry_Status	char(1)		Y		E	E=Entered, W=Withdrawn, D=Disqualified
Pit_Crew_Chief	varchar(60)					Name of the pit crew chief for this entry

Table 13: RaceResult

Stores the final race outcome for each race entry. One-to-one with RaceEntry.

Column Name	Data Type	Key	Mandatory	Indexed	Default Value	Remarks
RaceEntry_ID	uniqueidentifier	PK/ FK	Y	Y		References RaceEntry; one result per entry
Finish_Position	smallint					Final finishing position; NULL if DNF
Laps_Completed	smallint		Y		0	Total laps completed
Points_Earned	numeric(6,2)		Y		0.00	Championship points earned
Finish_Status	varchar(20)		Y			Classified, DNF, DNS, or DSQ
Prize_Won	numeric(12,2)				0.00	Prize money awarded in USD
Fastest_Lap_Time	numeric(8,3)					Fastest lap time in seconds

Section 3: Lab 2 - Physical Database

Lab 2 was me taking the design from Lab 1 and building it out in SQL. I wrote all the CREATE TABLE statements with the foreign key constraints and inserted test data into all the tables. I also fixed the relationship lines based on the Lab 1 feedback before writing any of this.

Section 3A: SQL Code

```
-- Apex Racing F1 Team Database
-- Vatsal Sagar
-- CIS 245 Lab 2
-- Montgomery County Community College
-- April 15, 2026

USE JCF_Student15_DB;
GO

-- drop tables first so we can rerun the script without errors
-- has to be in reverse order or the foreign keys will complain

IF OBJECT_ID('RaceResult', 'U') IS NOT NULL DROP TABLE RaceResult;
IF OBJECT_ID('RaceEntry', 'U') IS NOT NULL DROP TABLE RaceEntry;
IF OBJECT_ID('Race', 'U') IS NOT NULL DROP TABLE Race;
IF OBJECT_ID('Join_Team_Sponsor', 'U') IS NOT NULL DROP TABLE Join_Team_Sponsor;
IF OBJECT_ID('Sponsor', 'U') IS NOT NULL DROP TABLE Sponsor;
IF OBJECT_ID('Car', 'U') IS NOT NULL DROP TABLE Car;
IF OBJECT_ID('CarStatus_Lookup', 'U') IS NOT NULL DROP TABLE CarStatus_Lookup;
IF OBJECT_ID('CrewMember', 'U') IS NOT NULL DROP TABLE CrewMember;
IF OBJECT_ID('CrewRole_Lookup', 'U') IS NOT NULL DROP TABLE CrewRole_Lookup;
IF OBJECT_ID('Driver', 'U') IS NOT NULL DROP TABLE Driver;
IF OBJECT_ID('Track', 'U') IS NOT NULL DROP TABLE Track;
IF OBJECT_ID('Series_Lookup', 'U') IS NOT NULL DROP TABLE Series_Lookup;
IF OBJECT_ID('Team', 'U') IS NOT NULL DROP TABLE Team;
GO

-- create tables

CREATE TABLE Team (
    Team_ID UNIQUEIDENTIFIER NOT NULL DEFAULT NEWID(),
    Team_Name VARCHAR(60) NOT NULL,
    Team_BaseCity VARCHAR(40) NOT NULL,
    Team_BaseState VARCHAR(30) NOT NULL,
    Team_PrincipalName VARCHAR(60) NOT NULL,
    Team_Phone VARCHAR(20) NULL,
    Team_Email VARCHAR(80) NULL,
    Team_FoundedYear SMALLINT NOT NULL,
    Team_Status CHAR(1) NOT NULL DEFAULT 'A',
    CONSTRAINT PK_Team PRIMARY KEY (Team_ID)
);
GO
```

```
CREATE TABLE Driver (  
    Driver_ID UNIQUEIDENTIFIER NOT NULL DEFAULT NEWID(),  
    Team_ID UNIQUEIDENTIFIER NOT NULL,  
    Driver_FirstName VARCHAR(30) NOT NULL,  
    Driver_LastName VARCHAR(30) NOT NULL,  
    License_Number VARCHAR(25) NOT NULL,  
    DOB DATE NOT NULL,  
    Nationality VARCHAR(40) NOT NULL,  
    Experience_Years SMALLINT NULL DEFAULT 0,  
    Active_Flag BIT NOT NULL DEFAULT 1,  
    CONSTRAINT PK_Driver PRIMARY KEY (Driver_ID),  
    CONSTRAINT FK_Driver_Team FOREIGN KEY (Team_ID) REFERENCES Team(Team_ID)  
);  
GO  
  
CREATE TABLE CrewRole_Lookup (  
    CrewRole_Code CHAR(3) NOT NULL,  
    CrewRole_Description VARCHAR(40) NOT NULL,  
    CONSTRAINT PK_CrewRole_Lookup PRIMARY KEY (CrewRole_Code)  
);  
GO  
  
CREATE TABLE CrewMember (  
    CrewMember_ID UNIQUEIDENTIFIER NOT NULL DEFAULT NEWID(),  
    Team_ID UNIQUEIDENTIFIER NOT NULL,  
    CrewRole_Code CHAR(3) NOT NULL,  
    Crew_FirstName VARCHAR(30) NOT NULL,  
    Crew_LastName VARCHAR(30) NOT NULL,  
    Crew_Phone VARCHAR(20) NULL,  
    Crew_Email VARCHAR(80) NULL,  
    Hire_Date DATETIME NOT NULL,  
    Active_Flag BIT NOT NULL DEFAULT 1,  
    CONSTRAINT PK_CrewMember PRIMARY KEY (CrewMember_ID),  
    CONSTRAINT FK_CrewMember_Team FOREIGN KEY (Team_ID) REFERENCES Team(Team_ID),  
    CONSTRAINT FK_CrewMember_CrewRole FOREIGN KEY (CrewRole_Code) REFERENCES  
CrewRole_Lookup(CrewRole_Code)  
);  
GO  
  
CREATE TABLE CarStatus_Lookup (  
    CarStatus_Code CHAR(2) NOT NULL,  
    CarStatus_Description VARCHAR(30) NOT NULL,  
    CONSTRAINT PK_CarStatus_Lookup PRIMARY KEY (CarStatus_Code)  
);  
GO  
  
CREATE TABLE Car (  
    Car_ID UNIQUEIDENTIFIER NOT NULL DEFAULT NEWID(),  
    Team_ID UNIQUEIDENTIFIER NOT NULL,  
    CarStatus_Code CHAR(2) NOT NULL,  
    Car_Number VARCHAR(10) NOT NULL,  
    Chassis_Number VARCHAR(30) NOT NULL,  
    Engine_Supplier VARCHAR(40) NOT NULL,  
    Model_Name VARCHAR(40) NOT NULL,  
    Model_Year SMALLINT NOT NULL,  
    Last_Service_Date DATETIME NULL,  
    CONSTRAINT PK_Car PRIMARY KEY (Car_ID),  
    CONSTRAINT FK_Car_Team FOREIGN KEY (Team_ID) REFERENCES Team(Team_ID),  
    CONSTRAINT FK_Car_CarStatus FOREIGN KEY (CarStatus_Code) REFERENCES  
CarStatus_Lookup(CarStatus_Code)  
);  
GO
```

```
CREATE TABLE Sponsor (  
    Sponsor_ID UNIQUEIDENTIFIER NOT NULL DEFAULT NEWID(),  
    Sponsor_Name VARCHAR(80) NOT NULL,  
    Industry_Type VARCHAR(40) NOT NULL,  
    Contact_Name VARCHAR(30) NULL,  
    Contact_Email VARCHAR(80) NULL,  
    Contact_Phone VARCHAR(20) NULL,  
    Sponsor_Tier VARCHAR(20) NOT NULL,  
    Active_Flag BIT NOT NULL DEFAULT 1,  
    CONSTRAINT PK_Sponsor PRIMARY KEY (Sponsor_ID)  
);  
GO
```

```
CREATE TABLE Join_Team_Sponsor (  
    Team_ID UNIQUEIDENTIFIER NOT NULL,  
    Sponsor_ID UNIQUEIDENTIFIER NOT NULL,  
    Contract_StartDate DATETIME NOT NULL,  
    Contract_EndDate DATETIME NOT NULL,  
    Contribution_Amount NUMERIC(12,2) NOT NULL DEFAULT 0.00,  
    Primary_Sponsor_Flag BIT NOT NULL DEFAULT 0,  
    CONSTRAINT PK_Join_Team_Sponsor PRIMARY KEY (Team_ID, Sponsor_ID),  
    CONSTRAINT FK_JTS_Team FOREIGN KEY (Team_ID) REFERENCES Team(Team_ID),  
    CONSTRAINT FK_JTS_Sponsor FOREIGN KEY (Sponsor_ID) REFERENCES  
Sponsor(Sponsor_ID)  
);  
GO
```

```
CREATE TABLE Track (  
    Track_ID UNIQUEIDENTIFIER NOT NULL DEFAULT NEWID(),  
    Track_Name VARCHAR(80) NOT NULL,  
    Country_Name VARCHAR(40) NOT NULL,  
    City_Name VARCHAR(40) NOT NULL,  
    Track_Type VARCHAR(20) NOT NULL,  
    Length_Miles NUMERIC(5,2) NOT NULL,  
    Turns_Count SMALLINT NOT NULL,  
    FIA_Grade VARCHAR(10) NULL,  
    Active_Flag BIT NOT NULL DEFAULT 1,  
    CONSTRAINT PK_Track PRIMARY KEY (Track_ID)  
);  
GO
```

```
CREATE TABLE Series_Lookup (  
    Series_Code CHAR(4) NOT NULL,  
    Series_Description VARCHAR(40) NOT NULL,  
    CONSTRAINT PK_Series_Lookup PRIMARY KEY (Series_Code)  
);  
GO
```

```
CREATE TABLE Race (  
    Race_ID UNIQUEIDENTIFIER NOT NULL DEFAULT NEWID(),  
    Track_ID UNIQUEIDENTIFIER NOT NULL,  
    Series_Code CHAR(4) NOT NULL,  
    Race_Name VARCHAR(80) NOT NULL,  
    Season_Year SMALLINT NOT NULL,  
    Race_Date DATETIME NOT NULL,  
    Scheduled_Laps SMALLINT NOT NULL,  
    Weather_Summary VARCHAR(30) NULL,  
    Purse_Amount NUMERIC(12,2) NULL DEFAULT 0.00,  
    CONSTRAINT PK_Race PRIMARY KEY (Race_ID),
```

```

        CONSTRAINT FK_Race_Track FOREIGN KEY (Track_ID) REFERENCES Track(Track_ID),
        CONSTRAINT FK_Race_Series FOREIGN KEY (Series_Code) REFERENCES
Series_Lookup(Series_Code)
    );
GO

```

```

CREATE TABLE RaceEntry (
    RaceEntry_ID UNIQUEIDENTIFIER NOT NULL DEFAULT NEWID(),
    Race_ID UNIQUEIDENTIFIER NOT NULL,
    Car_ID UNIQUEIDENTIFIER NOT NULL,
    Driver_ID UNIQUEIDENTIFIER NOT NULL,
    Starting_Position SMALLINT NOT NULL,
    Qualifying_Time NUMERIC(8,3) NULL,
    Entry_Status CHAR(1) NOT NULL DEFAULT 'E',
    Pit_Crew_Chief VARCHAR(60) NULL,
    CONSTRAINT PK_RaceEntry PRIMARY KEY (RaceEntry_ID),
    CONSTRAINT FK_RaceEntry_Race FOREIGN KEY (Race_ID) REFERENCES Race(Race_ID),
    CONSTRAINT FK_RaceEntry_Car FOREIGN KEY (Car_ID) REFERENCES Car(Car_ID),
    CONSTRAINT FK_RaceEntry_Driver FOREIGN KEY (Driver_ID) REFERENCES
Driver(Driver_ID)
);
GO

```

```

CREATE TABLE RaceResult (
    RaceEntry_ID UNIQUEIDENTIFIER NOT NULL,
    Finish_Position SMALLINT NULL,
    Laps_Completed SMALLINT NOT NULL DEFAULT 0,
    Points_Earned NUMERIC(6,2) NOT NULL DEFAULT 0.00,
    Finish_Status VARCHAR(20) NOT NULL,
    Prize_Won NUMERIC(12,2) NULL DEFAULT 0.00,
    Fastest_Lap_Time NUMERIC(8,3) NULL,
    CONSTRAINT PK_RaceResult PRIMARY KEY (RaceEntry_ID),
    CONSTRAINT FK_RaceResult_RaceEntry FOREIGN KEY (RaceEntry_ID) REFERENCES
RaceEntry(RaceEntry_ID)
);
GO

```

```

-- insert test data
-- SQL auto-generates all IDs using NEWID() so we dont need to write them out
-- for tables that reference other tables we use a variable or subquery to grab the ID

```

```

-- crew roles
INSERT INTO CrewRole_Lookup (CrewRole_Code, CrewRole_Description) VALUES
('ENG', 'Race Engineer'),
('MEC', 'Mechanic'),
('CHF', 'Chief Mechanic'),
('STR', 'Strategist'),
('DAT', 'Data Analyst'),
('PHY', 'Physiotherapist'),
('NUT', 'Nutritionist'),
('PRE', 'Press Officer'),
('LOG', 'Logistics Manager'),
('QUA', 'Quality Control');
GO

```

```

-- car statuses
INSERT INTO CarStatus_Lookup (CarStatus_Code, CarStatus_Description) VALUES
('AC', 'Active'),
('SP', 'Spare'),
('DC', 'Decommissioned'),
('RP', 'Under Repair');

```

GO

```
-- racing series
```

```
INSERT INTO Series_Lookup (Series_Code, Series_Description) VALUES
```

```
('F1 ', 'Formula 1 World Championship'),
```

```
('F2 ', 'Formula 2 Championship'),
```

```
('F3 ', 'Formula 3 Championship'),
```

```
('FE ', 'Formula E World Championship');
```

GO

```
-- team (ID gets auto generated by SQL)
```

```
INSERT INTO Team (Team_Name, Team_BaseCity, Team_BaseState, Team_PrincipalName,
```

```
Team_Phone, Team_Email, Team_FoundedYear, Team_Status)
```

```
VALUES ('Apex Racing', 'Silverstone', 'Northamptonshire', 'James Whitfield', '+44-1234-567890', 'info@apexracing.com', 2010, 'A');
```

GO

```
-- grab the team ID so we can use it below
```

```
DECLARE @TeamID UNIQUEIDENTIFIER = (SELECT Team_ID FROM Team WHERE Team_Name = 'Apex Racing');
```

```
-- drivers
```

```
INSERT INTO Driver (Team_ID, Driver_FirstName, Driver_LastName, License_Number, DOB, Nationality, Experience_Years, Active_Flag) VALUES
```

```
(@TeamID, 'Liam', 'Carter', 'LIC-001', '1995-03-12', 'British', 8, 1),
```

```
(@TeamID, 'Marco', 'Rossi', 'LIC-002', '1997-07-24', 'Italian', 6, 1),
```

```
(@TeamID, 'Sofia', 'Almeida', 'LIC-003', '1999-11-05', 'Brazilian', 4, 1),
```

```
(@TeamID, 'Riku', 'Tanaka', 'LIC-004', '1996-01-18', 'Japanese', 7, 1),
```

```
(@TeamID, 'Erik', 'Lindqvist', 'LIC-005', '1998-09-30', 'Swedish', 5, 1),
```

```
(@TeamID, 'Carlos', 'Vega', 'LIC-006', '2000-04-14', 'Spanish', 3, 1),
```

```
(@TeamID, 'Aiden', 'Murphy', 'LIC-007', '1994-06-22', 'Irish', 9, 1),
```

```
(@TeamID, 'Priya', 'Sharma', 'LIC-008', '2001-02-08', 'Indian', 2, 1),
```

```
(@TeamID, 'Felix', 'Braun', 'LIC-009', '1993-12-01', 'German', 10, 0),
```

```
(@TeamID, 'Yusuf', 'Hassan', 'LIC-010', '2002-08-17', 'Moroccan', 1, 1),
```

```
(@TeamID, 'Natalie', 'Dubois', 'LIC-011', '1997-05-29', 'French', 6, 1),
```

```
(@TeamID, 'Anton', 'Petrov', 'LIC-012', '1999-10-03', 'Russian', 4, 1),
```

```
(@TeamID, 'Jake', 'Williams', 'LIC-013', '1995-07-11', 'American', 8, 1),
```

```
(@TeamID, 'Hana', 'Kovac', 'LIC-014', '2000-03-25', 'Czech', 3, 1),
```

```
(@TeamID, 'Oscar', 'Bennett', 'LIC-015', '1998-11-19', 'Australian', 5, 1),
```

```
(@TeamID, 'Mei', 'Zhang', 'LIC-016', '2001-06-07', 'Chinese', 2, 1),
```

```
(@TeamID, 'Diego', 'Herrera', 'LIC-017', '1996-09-14', 'Mexican', 7, 1),
```

```
(@TeamID, 'Sven', 'Eriksson', 'LIC-018', '1994-04-28', 'Norwegian', 9, 0),
```

```
(@TeamID, 'Amara', 'Osei', 'LIC-019', '2003-01-15', 'Ghanaian', 0, 1),
```

```
(@TeamID, 'Tomas', 'Novak', 'LIC-020', '1997-08-20', 'Slovak', 6, 1);
```

GO

```
-- crew members
```

```
DECLARE @TeamID2 UNIQUEIDENTIFIER = (SELECT Team_ID FROM Team WHERE Team_Name = 'Apex Racing');
```

```
INSERT INTO CrewMember (Team_ID, CrewRole_Code, Crew_FirstName, Crew_LastName,
```

```
Crew_Phone, Crew_Email, Hire_Date, Active_Flag) VALUES
```

```
(@TeamID2, 'CHF', 'David', 'Knight', '+44-7700-100001', 'dknight@apexracing.com',
```

```
'2010-06-01', 1),
```

```
(@TeamID2, 'ENG', 'Sarah', 'Cole', '+44-7700-100028', 'scole@apexracing.com', '2012-
```

```
03-15', 1),
```

```
(@TeamID2, 'ENG', 'Tom', 'Harris', '+44-7700-100034', 'tharris@apexracing.com', '2013-
```

```
11-03', 1),
```

```
(@TeamID2, 'MEC', 'Phil', 'Owen', '+44-7700-100057', 'powen@apexracing.com', '2011-09-
```

```
10', 1),
```

```
(@TeamID2, 'MEC', 'Lisa', 'Ward', '+44-7700-100062', 'lward@apexracing.com', '2014-02-28', 1),
(@TeamID2, 'MEC', 'Ben', 'Fox', '+44-7700-100089', 'bfox@apexracing.com', '2015-07-19', 1),
(@TeamID2, 'STR', 'Claire', 'Stone', '+44-7700-100091', 'cstone@apexracing.com', '2016-08-01', 1),
(@TeamID2, 'STR', 'James', 'Hunt', '+44-7700-100103', 'jhunt@apexracing.com', '2017-04-11', 1),
(@TeamID2, 'DAT', 'Nina', 'Brooks', '+44-7700-100115', 'nbrooks@apexracing.com', '2018-04-30', 1),
(@TeamID2, 'DAT', 'Alex', 'Turner', '+44-7700-100122', 'aturner@apexracing.com', '2019-10-07', 1),
(@TeamID2, 'PHY', 'Emma', 'Lawson', '+44-7700-100138', 'elawson@apexracing.com', '2013-11-05', 1),
(@TeamID2, 'NUT', 'Oliver', 'Green', '+44-7700-100141', 'ogreen@apexracing.com', '2020-02-14', 1),
(@TeamID2, 'PRE', 'Julia', 'Adams', '+44-7700-100155', 'jadams@apexracing.com', '2011-07-19', 1),
(@TeamID2, 'LOG', 'Mark', 'Taylor', '+44-7700-100167', 'mtaylor@apexracing.com', '2010-09-01', 1),
(@TeamID2, 'QUA', 'Helen', 'Price', '+44-7700-100178', 'hprice@apexracing.com', '2012-12-03', 1),
(@TeamID2, 'MEC', 'Chris', 'Evans', '+44-7700-100184', 'cevans@apexracing.com', '2021-03-17', 1),
(@TeamID2, 'ENG', 'Zoe', 'Mitchell', '+44-7700-100196', 'zmitchell@apexracing.com', '2022-01-10', 1),
(@TeamID2, 'STR', 'Ryan', 'Clark', '+44-7700-100207', 'rclark@apexracing.com', '2023-08-14', 1),
(@TeamID2, 'DAT', 'Chloe', 'Robinson', '+44-7700-100219', 'crobinson@apexracing.com', '2024-07-08', 0),
(@TeamID2, 'MEC', 'George', 'Wilson', '+44-7700-100231', 'gwilson@apexracing.com', '2025-01-14', 1);
GO
```

```
-- cars
```

```
DECLARE @TeamID3 UNIQUEIDENTIFIER = (SELECT Team_ID FROM Team WHERE Team_Name = 'Apex Racing');
```

```
INSERT INTO Car (Team_ID, CarStatus_Code, Car_Number, Chassis_Number, Engine_Supplier, Model_Name, Model_Year, Last_Service_Date) VALUES
(@TeamID3, 'AC', '1', 'APEX-CH-2024-001', 'Mercedes-AMG', 'Apex A24', 2024, '2024-02-10'),
(@TeamID3, 'AC', '2', 'APEX-CH-2024-002', 'Mercedes-AMG', 'Apex A24', 2024, '2024-02-12'),
(@TeamID3, 'SP', '3', 'APEX-CH-2024-003', 'Mercedes-AMG', 'Apex A24', 2024, '2024-01-20'),
(@TeamID3, 'AC', '4', 'APEX-CH-2023-001', 'Mercedes-AMG', 'Apex A23', 2023, '2023-11-15'),
(@TeamID3, 'DC', '5', 'APEX-CH-2022-001', 'Mercedes-AMG', 'Apex A22', 2022, '2023-01-05'),
(@TeamID3, 'AC', '6', 'APEX-CH-2025-001', 'Ferrari PU', 'Apex A25', 2025, '2025-01-18'),
(@TeamID3, 'AC', '7', 'APEX-CH-2025-002', 'Ferrari PU', 'Apex A25', 2025, '2025-01-20'),
(@TeamID3, 'SP', '8', 'APEX-CH-2025-003', 'Ferrari PU', 'Apex A25', 2025, '2025-02-01'),
(@TeamID3, 'RP', '9', 'APEX-CH-2023-002', 'Mercedes-AMG', 'Apex A23', 2023, '2024-08-30'),
(@TeamID3, 'AC', '10', 'APEX-CH-2025-004', 'Ferrari PU', 'Apex A25', 2025, '2025-02-10'),
(@TeamID3, 'DC', '11', 'APEX-CH-2021-001', 'Honda RA621H', 'Apex A21', 2021, '2022-06-12'),
```

```
(@TeamID3, 'AC', '12', 'APEX-CH-2025-005', 'Ferrari PU', 'Apex A25', 2025, '2025-02-14'),
(@TeamID3, 'SP', '13', 'APEX-CH-2024-004', 'Mercedes-AMG', 'Apex A24', 2024, '2024-03-05'),
(@TeamID3, 'AC', '14', 'APEX-CH-2025-006', 'Ferrari PU', 'Apex A25', 2025, '2025-03-01'),
(@TeamID3, 'RP', '15', 'APEX-CH-2024-005', 'Mercedes-AMG', 'Apex A24', 2024, '2024-09-18'),
(@TeamID3, 'AC', '16', 'APEX-CH-2025-007', 'Ferrari PU', 'Apex A25', 2025, '2025-03-10'),
(@TeamID3, 'DC', '17', 'APEX-CH-2020-001', 'Renault ET20', 'Apex A20', 2020, '2021-11-01'),
(@TeamID3, 'SP', '18', 'APEX-CH-2025-008', 'Ferrari PU', 'Apex A25', 2025, '2025-02-28'),
(@TeamID3, 'AC', '19', 'APEX-CH-2025-009', 'Ferrari PU', 'Apex A25', 2025, '2025-03-15'),
(@TeamID3, 'RP', '20', 'APEX-CH-2023-003', 'Mercedes-AMG', 'Apex A23', 2023, '2024-07-22');
GO
```

```
-- sponsors (SQL generates the IDs automatically)
INSERT INTO Sponsor (Sponsor_Name, Industry_Type, Contact_Name, Contact_Email,
Contact_Phone, Sponsor_Tier, Active_Flag) VALUES
('RedLine Energy', 'Energy Drinks', 'Mike Ford', 'mford@redline.com', '+1-800-111-0001', 'Title', 1),
('TurboTech Systems', 'Technology', 'Anna Lee', 'alee@turbotech.com', '+1-800-111-0002', 'Primary', 1),
('FastFuel Ltd', 'Petroleum', 'Steve King', 'sking@fastfuel.com', '+44-800-222-0003', 'Primary', 1),
('Apex Tyres Co', 'Manufacturing', 'Lisa Park', 'lpark@apextyres.com', '+49-800-333-0004', 'Associate', 1),
('VelocityBank', 'Finance', 'Tom Marsh', 'tmarsh@velbank.com', '+1-800-111-0005', 'Associate', 1),
('Omega Insurance', 'Insurance', 'Jane Grey', 'jgrey@omegains.com', '+44-800-222-0006', 'Associate', 1),
('Circuit Apparel', 'Fashion', 'Paulo Mota', 'pmota@capparel.com', '+55-800-444-0007', 'Associate', 1),
('PitStop Logistics', 'Logistics', 'Raj Patel', 'rpatel@pitstop.com', '+91-800-555-0008', 'Associate', 1),
('NanoCoat Materials', 'Materials', 'Clara Beck', 'cbeck@nanocoat.com', '+49-800-333-0009', 'Associate', 0),
('GridComm Wireless', 'Telecom', 'Dan Yung', 'dyung@gridcomm.com', '+1-800-111-0010', 'Primary', 1),
('SkyView Media', 'Media', 'Sophie Hart', 'shart@skyview.com', '+44-800-222-0011', 'Associate', 1),
('ProDrive Tools', 'Tools', 'Ivan Roth', 'iroth@prodrive.com', '+49-800-333-0012', 'Associate', 1),
('AeroParts Global', 'Aerospace', 'Fiona Chan', 'fchan@aeroparts.com', '+86-800-666-0013', 'Associate', 1),
('NextGen Composites', 'Materials', 'Eric Dahl', 'edahl@nextgen.com', '+46-800-777-0014', 'Associate', 1),
('ClearVision Optics', 'Technology', 'Maria Santos', 'msantos@clearvision.com', '+351-800-888-0015', 'Associate', 1),
('SpeedBet Wagering', 'Gaming', 'Alan Ross', 'aross@speedbet.com', '+44-800-222-0016', 'Associate', 0),
('FuelCell Energy', 'Energy', 'Kim Ahn', 'kahn@fuelcell.com', '+82-800-999-0017', 'Associate', 1),
('ThrottleWear', 'Apparel', 'Luis Mora', 'lmora@throttlewear.com', '+52-800-100-0018', 'Associate', 1),
('PeakPerformance Nutrition', 'Health', 'Gina Wolfe', 'gwolfe@peakperf.com', '+1-800-111-0019', 'Associate', 1),
('AlpineAir Travel', 'Travel', 'Hector Ruiz', 'hruiz@alpineair.com', '+34-800-200-0020', 'Associate', 1);
```


GO

```
-- link sponsors to team using subqueries to look up the IDs by name
DECLARE @TID UNIQUEIDENTIFIER = (SELECT Team_ID FROM Team WHERE Team_Name = 'Apex
Racing');
```

```
INSERT INTO Join_Team_Sponsor (Team_ID, Sponsor_ID, Contract_StartDate,
Contract_EndDate, Contribution_Amount, Primary_Sponsor_Flag) VALUES
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'RedLine Energy'),
'2024-01-01', '2026-12-31', 25000000.00, 1),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'TurboTech Systems'),
'2024-01-01', '2025-12-31', 11750000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'FastFuel Ltd'),
'2023-01-01', '2025-12-31', 9800000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'Apex Tyres Co'),
'2024-03-01', '2026-02-28', 4250000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'VelocityBank'),
'2023-06-01', '2025-05-31', 3175000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'Omega Insurance'),
'2022-01-01', '2024-12-31', 2000000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'Circuit Apparel'),
'2024-05-01', '2026-04-30', 1625000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'PitStop Logistics'),
'2023-01-01', '2025-12-31', 980000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'NanoCoat Materials'),
'2021-01-01', '2023-12-31', 750000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'GridComm Wireless'),
'2024-01-01', '2026-12-31', 8500000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'SkyView Media'),
'2024-06-01', '2026-05-31', 1350000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'ProDrive Tools'),
'2023-09-01', '2025-08-31', 725000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'AeroParts Global'),
'2024-02-01', '2026-01-31', 2250000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'NextGen Composites'),
'2023-04-01', '2025-03-31', 580000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'ClearVision Optics'),
'2025-01-01', '2026-12-31', 875000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'SpeedBet Wagering'),
'2022-06-01', '2024-05-31', 400000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'FuelCell Energy'),
'2024-07-01', '2026-06-30', 1100000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'ThrottleWear'),
'2025-01-01', '2026-12-31', 475000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'PeakPerformance
Nutrition'),'2023-11-01', '2025-10-31', 615000.00, 0),
(@TID, (SELECT Sponsor_ID FROM Sponsor WHERE Sponsor_Name = 'AlpineAir Travel'),
'2024-08-01', '2026-07-31', 320000.00, 0);
GO
```

```
-- tracks
INSERT INTO Track (Track_Name, Country_Name, City_Name, Track_Type, Length_Miles,
Turns_Count, FIA_Grade, Active_Flag) VALUES
('Silverstone Circuit', 'United Kingdom', 'Silverstone', 'Road Course', 3.66, 18,
'Grade 1', 1),
('Circuit de Monaco', 'Monaco', 'Monte Carlo', 'Street Circuit', 2.07, 19, 'Grade 1',
1),
('Autodromo Nazionale Monza', 'Italy', 'Monza', 'Road Course', 3.60, 11, 'Grade 1',
1),
('Circuit of the Americas', 'USA', 'Austin', 'Road Course', 3.41, 20, 'Grade 1', 1),
('Suzuka International Circuit', 'Japan', 'Suzuka', 'Road Course', 3.61, 18, 'Grade
1', 1),
```

```
( 'Spa-Francorchamps', 'Belgium', 'Stavelot', 'Road Course', 4.35, 19, 'Grade 1', 1),
( 'Circuit Gilles Villeneuve', 'Canada', 'Montreal', 'Street Circuit', 2.71, 14, 'Grade 1', 1),
( 'Hungaroring', 'Hungary', 'Budapest', 'Road Course', 2.72, 14, 'Grade 1', 1),
( 'Red Bull Ring', 'Austria', 'Spielberg', 'Road Course', 2.68, 9, 'Grade 1', 1),
( 'Bahrain International Circuit', 'Bahrain', 'Sakhir', 'Road Course', 3.36, 15, 'Grade 1', 1),
( 'Yas Marina Circuit', 'UAE', 'Abu Dhabi', 'Road Course', 3.28, 16, 'Grade 1', 1),
( 'Jeddah Corniche Circuit', 'Saudi Arabia', 'Jeddah', 'Street Circuit', 3.83, 27, 'Grade 1', 1),
( 'Melbourne Grand Prix Circuit', 'Australia', 'Melbourne', 'Street Circuit', 3.30, 16, 'Grade 1', 1),
( 'Circuit de Barcelona-Catalunya', 'Spain', 'Barcelona', 'Road Course', 2.89, 16, 'Grade 1', 1),
( 'Autodromo Hermanos Rodriguez', 'Mexico', 'Mexico City', 'Road Course', 2.67, 17, 'Grade 1', 1),
( 'Interlagos Circuit', 'Brazil', 'Sao Paulo', 'Road Course', 2.68, 15, 'Grade 1', 1),
( 'Marina Bay Street Circuit', 'Singapore', 'Singapore', 'Street Circuit', 3.15, 23, 'Grade 1', 1),
( 'Losail International Circuit', 'Qatar', 'Lusail', 'Road Course', 3.37, 16, 'Grade 1', 1),
( 'Las Vegas Street Circuit', 'USA', 'Las Vegas', 'Street Circuit', 3.86, 17, 'Grade 1', 1),
( 'Zandvoort Circuit', 'Netherlands', 'Zandvoort', 'Road Course', 2.65, 14, 'Grade 1', 1);
GO
```

```
-- races - look up track IDs by name
```

```
INSERT INTO Race (Track_ID, Series_Code, Race_Name, Season_Year, Race_Date,
Scheduled_Laps, Weather_Summary, Purse_Amount) VALUES
((SELECT Track_ID FROM Track WHERE Track_Name = 'Bahrain International Circuit'),
'F1 ', 'Bahrain Grand Prix', 2025, '2025-03-02 15:00:00', 57, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Jeddah Corniche Circuit'),
'F1 ', 'Saudi Arabian Grand Prix', 2025, '2025-03-16 20:00:00', 50, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Melbourne Grand Prix Circuit'),
'F1 ', 'Australian Grand Prix', 2025, '2025-03-30 15:00:00', 58, 'Mixed',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Suzuka International Circuit'),
'F1 ', 'Japanese Grand Prix', 2025, '2025-04-06 14:00:00', 53, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Circuit de Barcelona-Catalunya'), 'F1 ',
'Spanish Grand Prix', 2025, '2025-05-25 15:00:00', 66, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Circuit de Monaco'), 'F1 ',
'Monaco Grand Prix', 2025, '2025-05-25 15:00:00', 78, 'Wet',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Circuit Gilles Villeneuve'), 'F1 ',
'Canadian Grand Prix', 2025, '2025-06-15 18:00:00', 70, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Red Bull Ring'), 'F1 ',
'Austrian Grand Prix', 2025, '2025-06-29 15:00:00', 71, 'Mixed',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Silverstone Circuit'), 'F1 ',
'British Grand Prix', 2025, '2025-07-06 15:00:00', 52, 'Wet',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Hungaroring'), 'F1 ',
'Hungarian Grand Prix', 2025, '2025-07-27 15:00:00', 70, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Spa-Francorchamps'), 'F1 ',
'Belgian Grand Prix', 2025, '2025-08-24 15:00:00', 44, 'Wet',
25000000.00),
```

```

((SELECT Track_ID FROM Track WHERE Track_Name = 'Zandvoort Circuit'),      'F1
', 'Dutch Grand Prix',          2025, '2025-08-31 15:00:00', 72, 'Mixed',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Autodromo Nazionale Monza'), 'F1
', 'Italian Grand Prix',        2025, '2025-09-07 15:00:00', 53, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Marina Bay Street Circuit'), 'F1
', 'Singapore Grand Prix',      2025, '2025-09-21 20:00:00', 62, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Losail International Circuit'), 'F1
', 'Qatar Grand Prix',          2025, '2025-10-05 19:00:00', 57, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Circuit of the Americas'), 'F1
', 'United States Grand Prix',  2025, '2025-10-19 19:00:00', 56, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Autodromo Hermanos Rodriguez'), 'F1
', 'Mexico City Grand Prix',    2025, '2025-10-26 20:00:00', 71, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Interlagos Circuit'),      'F1
', 'Sao Paulo Grand Prix',      2025, '2025-11-09 17:00:00', 71, 'Wet',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Las Vegas Street Circuit'), 'F1
', 'Las Vegas Grand Prix',      2025, '2025-11-22 22:00:00', 50, 'Dry',
25000000.00),
((SELECT Track_ID FROM Track WHERE Track_Name = 'Yas Marina Circuit'),      'F1
', 'Abu Dhabi Grand Prix',      2025, '2025-12-07 17:00:00', 58, 'Dry',
25000000.00);
GO

```

-- race entries - look up IDs by name/number/license

```

DECLARE @Car1 UNIQUEIDENTIFIER = (SELECT Car_ID FROM Car WHERE Car_Number = '1');
DECLARE @Car2 UNIQUEIDENTIFIER = (SELECT Car_ID FROM Car WHERE Car_Number = '2');
DECLARE @Drv1 UNIQUEIDENTIFIER = (SELECT Driver_ID FROM Driver WHERE License_Number =
'LIC-001');
DECLARE @Drv2 UNIQUEIDENTIFIER = (SELECT Driver_ID FROM Driver WHERE License_Number =
'LIC-002');

```

```

INSERT INTO RaceEntry (Race_ID, Car_ID, Driver_ID, Starting_Position, Qualifying_Time,
Entry_Status, Pit_Crew_Chief) VALUES
((SELECT Race_ID FROM Race WHERE Race_Name = 'Bahrain Grand Prix'),      @Car1,
@Drv1,  3, 91.234, 'E', 'David Knight'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Bahrain Grand Prix'),      @Car2,
@Drv2,  7, 91.876, 'E', 'Tom Harris'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Saudi Arabian Grand Prix'), @Car1,
@Drv1,  2, 89.512, 'E', 'David Knight'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Saudi Arabian Grand Prix'), @Car2,
@Drv2,  5, 90.003, 'E', 'Tom Harris'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Australian Grand Prix'),   @Car1,
@Drv1,  1, 85.456, 'E', 'David Knight'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Australian Grand Prix'),   @Car2,
@Drv2,  4, 86.123, 'E', 'Sarah Cole'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Japanese Grand Prix'),     @Car1,
@Drv1,  6, 88.901, 'E', 'David Knight'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Japanese Grand Prix'),     @Car2,
@Drv2,  9, 89.445, 'E', 'Tom Harris'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Spanish Grand Prix'),      @Car1,
@Drv1,  4, 79.112, 'E', 'David Knight'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Spanish Grand Prix'),      @Car2,
@Drv2,  8, 79.987, 'E', 'Sarah Cole'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Monaco Grand Prix'),      @Car1,
@Drv1,  2, 74.234, 'E', 'David Knight'),

```

```

((SELECT Race_ID FROM Race WHERE Race_Name = 'Monaco Grand Prix'),      @Car2,
@Drv2, 6, 74.887, 'E', 'Tom Harris'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Canadian Grand Prix'),    @Car1,
@Drv1, 3, 72.331, 'E', 'David Knight'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Canadian Grand Prix'),    @Car2,
@Drv2, 11, 73.002, 'E', 'Sarah Cole'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Austrian Grand Prix'),    @Car1,
@Drv1, 5, 64.123, 'E', 'David Knight'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Austrian Grand Prix'),    @Car2,
@Drv2, 10, 64.788, 'E', 'Tom Harris'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'British Grand Prix'),     @Car1,
@Drv1, 2, 88.234, 'E', 'David Knight'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'British Grand Prix'),     @Car2,
@Drv2, 7, 89.001, 'E', 'Tom Harris'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Hungarian Grand Prix'),   @Car1,
@Drv1, 4, 79.456, 'W', 'David Knight'),
((SELECT Race_ID FROM Race WHERE Race_Name = 'Hungarian Grand Prix'),   @Car2,
@Drv2, 12, 80.123, 'E', 'Sarah Cole');
GO

```

```

-- race results - look up RaceEntry by race name + driver license
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 2, 57, 18.00, 'Classified', 800000.00, 91.102 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Bahrain Grand Prix' AND D.License_Number = 'LIC-001';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 8, 57, 4.00, 'Classified', 200000.00, 91.990 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Bahrain Grand Prix' AND D.License_Number = 'LIC-002';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 1, 50, 25.00, 'Classified', 1200000.00, 89.234 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Saudi Arabian Grand Prix' AND D.License_Number = 'LIC-001';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 5, 50, 10.00, 'Classified', 400000.00, 90.112 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Saudi Arabian Grand Prix' AND D.License_Number = 'LIC-002';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 1, 58, 25.00, 'Classified', 1200000.00, 85.112 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Australian Grand Prix' AND D.License_Number = 'LIC-001';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 4, 58, 12.00, 'Classified', 500000.00, 86.334 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Australian Grand Prix' AND D.License_Number = 'LIC-002';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 6, 53, 8.00, 'Classified', 300000.00, 89.001 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Japanese Grand Prix' AND D.License_Number = 'LIC-001';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, NULL, 12, 0.00, 'DNF', 0.00, NULL FROM
RaceEntry RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID =
D.Driver_ID WHERE R.Race_Name = 'Japanese Grand Prix' AND D.License_Number =
'LIC-002';

```

```

INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 3, 66, 15.00, 'Classified', 600000.00, 79.223 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Spanish Grand Prix' AND D.License_Number = 'LIC-001';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 10, 66, 1.00, 'Classified', 100000.00, 80.456 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Spanish Grand Prix' AND D.License_Number = 'LIC-002';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 2, 78, 18.00, 'Classified', 800000.00, 74.112 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Monaco Grand Prix' AND D.License_Number = 'LIC-001';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 7, 78, 6.00, 'Classified', 250000.00, 75.003 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Monaco Grand Prix' AND D.License_Number = 'LIC-002';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 1, 70, 25.00, 'Classified', 1200000.00, 72.009 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Canadian Grand Prix' AND D.License_Number = 'LIC-001';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 9, 70, 2.00, 'Classified', 150000.00, 73.445 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Canadian Grand Prix' AND D.License_Number = 'LIC-002';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 4, 71, 12.00, 'Classified', 500000.00, 64.556 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Austrian Grand Prix' AND D.License_Number = 'LIC-001';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, NULL, 0, 0.00, 'DNS', 0.00, NULL FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Austrian Grand Prix' AND D.License_Number = 'LIC-002';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 3, 52, 15.00, 'Classified', 600000.00, 88.001 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'British Grand Prix' AND D.License_Number = 'LIC-001';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 6, 52, 8.00, 'Classified', 300000.00, 89.223 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'British Grand Prix' AND D.License_Number = 'LIC-002';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, NULL, 0, 0.00, 'DNS', 0.00, NULL FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Hungarian Grand Prix' AND D.License_Number = 'LIC-001';
INSERT INTO RaceResult (RaceEntry_ID, Finish_Position, Laps_Completed, Points_Earned,
Finish_Status, Prize_Won, Fastest_Lap_Time)
SELECT RE.RaceEntry_ID, 5, 70, 10.00, 'Classified', 400000.00, 80.334 FROM RaceEntry
RE JOIN Race R ON RE.Race_ID = R.Race_ID JOIN Driver D ON RE.Driver_ID = D.Driver_ID
WHERE R.Race_Name = 'Hungarian Grand Prix' AND D.License_Number = 'LIC-002';
GO

PRINT 'Apex Racing database setup complete.';

```

GO

Section 4: Lab 3 - Views

Lab 3 was writing 8 SQL views against the database from Lab 2. Each view is a different query type as laid out in the lab requirements.

Section 4A: Drop Existing Views

```
IF OBJECT_ID('vw_RaceEntryFullDetail', 'V') IS NOT NULL DROP VIEW
vw_RaceEntryFullDetail;
IF OBJECT_ID('vw_ClassifiedResults', 'V') IS NOT NULL DROP VIEW vw_ClassifiedResults;
IF OBJECT_ID('vw_AllRacesWithTrackAndSeries', 'V') IS NOT NULL DROP VIEW
vw_AllRacesWithTrackAndSeries;
IF OBJECT_ID('vw_DriversWithPoints', 'V') IS NOT NULL DROP VIEW vw_DriversWithPoints;
IF OBJECT_ID('vw_ActiveCars', 'V') IS NOT NULL DROP VIEW vw_ActiveCars;
IF OBJECT_ID('vw_AllDriversWithTeam', 'V') IS NOT NULL DROP VIEW
vw_AllDriversWithTeam;
IF OBJECT_ID('vw_ActiveSponsors', 'V') IS NOT NULL DROP VIEW vw_ActiveSponsors;
IF OBJECT_ID('vw_AllTracks', 'V') IS NOT NULL DROP VIEW vw_AllTracks;
GO
```

Section 4B: View 1: Race Entry Full Detail (4 Tables, WHERE, ORDER BY)

Pulls from RaceEntry, Driver, Car, and Race. Filters to confirmed entries where Entry_Status is E and orders by race date then starting position.

```
USE JCF_Student15_DB;
GO

CREATE VIEW vw_RaceEntryFullDetail AS
SELECT TOP 100 PERCENT
    D.Driver_FirstName,
    D.Driver_LastName,
    D.License_Number,
    C.Car_Number,
    C.Model_Name,
    R.Race_Name,
    R.Race_Date,
    RE.Starting_Position,
    RE.Qualifying_Time,
    RE.Entry_Status,
    RE.Pit_Crew_Chief
FROM RaceEntry RE
INNER JOIN Driver D ON RE.Driver_ID = D.Driver_ID
INNER JOIN Car C ON RE.Car_ID = C.Car_ID
INNER JOIN Race R ON RE.Race_ID = R.Race_ID
```

```
WHERE RE.Entry_Status = 'E'
ORDER BY R.Race_Date, RE.Starting_Position;
GO
```

Section 4C: View 2: Classified Race Results (3 Tables, WHERE, INNER JOIN)

Joins RaceResult, RaceEntry, and Race using INNER JOINS. Filters to only classified finishers, excluding DNF and DNS entries.

```
USE JCF_Student15_DB;
GO

CREATE VIEW vw_ClassifiedResults AS
SELECT
    RR.Finish_Position,
    RR.Laps_Completed,
    RR.Points_Earned,
    RR.Prize_Won,
    RR.Fastest_Lap_Time,
    RE.Starting_Position,
    RE.Qualifying_Time,
    RE.Pit_Crew_Chief,
    R.Race_Name,
    R.Season_Year,
    R.Race_Date
FROM RaceResult RR
INNER JOIN RaceEntry RE ON RR.RaceEntry_ID = RE.RaceEntry_ID
INNER JOIN Race R ON RE.Race_ID = R.Race_ID
WHERE RR.Finish_Status = 'Classified';
GO
```

Section 4D: View 3: All Races with Track and Series Info (3 Tables, No WHERE)

Joins Race, Track, and Series_Lookup to show every race on the calendar. No filter applied so all races come back.

```
USE JCF_Student15_DB;
GO

CREATE VIEW vw_AllRacesWithTrackAndSeries AS
SELECT
    R.Race_Name,
    R.Season_Year,
    R.Race_Date,
    R.Scheduled_Laps,
    R.Weather_Summary,
    R.Purse_Amount,
    T.Track_Name,
```

```

        T.Country_Name,
        T.City_Name,
        T.Track_Type,
        T.Length_Miles,
        S.Series_Description
FROM Race R
INNER JOIN Track T ON R.Track_ID = T.Track_ID
INNER JOIN Series_Lookup S ON R.Series_Code = S.Series_Code;
GO

```

Section 4E: View 4: Drivers Who Have Scored Points (Nested Query, 3 Tables)

Uses a nested subquery across Driver, RaceEntry, and RaceResult. The inner query finds driver IDs with at least one result above zero points and the outer query returns those drivers.

```

USE JCF_Student15_DB;
GO

CREATE VIEW vw_DriversWithPoints AS
SELECT
    D.Driver_FirstName,
    D.Driver_LastName,
    D.Nationality,
    D.License_Number,
    D.Experience_Years
FROM Driver D
WHERE D.Driver_ID IN (
    SELECT RE.Driver_ID
    FROM RaceEntry RE
    INNER JOIN RaceResult RR ON RE.RaceEntry_ID = RR.RaceEntry_ID
    WHERE RR.Points_Earned > 0
);
GO

```

Section 4F: View 5: Active Cars (2 Tables, WHERE)

Joins Car and CarStatus_Lookup and filters to only cars with status code AC.

```

USE JCF_Student15_DB;
GO

CREATE VIEW vw_ActiveCars AS
SELECT
    C.Car_Number,
    C.Chassis_Number,
    C.Engine_Supplier,
    C.Model_Name,
    C.Model_Year,

```



```
        C.Last_Service_Date,
        CS.CarStatus_Description
FROM Car C
INNER JOIN CarStatus_Lookup CS ON C.CarStatus_Code = CS.CarStatus_Code
WHERE C.CarStatus_Code = 'AC';
GO
```

Section 4G: View 6: All Drivers with Team (2 Tables, No WHERE, OUTER JOIN)

Uses a LEFT OUTER JOIN between Team and Driver so the team row always shows up even without drivers attached. No filter applied.

```
USE JCF_Student15_DB;
GO

CREATE VIEW vw_AllDriversWithTeam AS
SELECT
    T.Team_Name,
    T.Team_BaseCity,
    D.Driver_FirstName,
    D.Driver_LastName,
    D.Nationality,
    D.License_Number,
    D.Experience_Years,
    D.Active_Flag
FROM Team T
LEFT OUTER JOIN Driver D ON T.Team_ID = D.Team_ID;
GO
```

Section 4H: View 7: Active Sponsors (1 Table, WHERE)

Queries the Sponsor table filtered to only sponsors where Active_Flag is 1.

```
USE JCF_Student15_DB;
GO

CREATE VIEW vw_ActiveSponsors AS
SELECT
    Sponsor_Name,
    Industry_Type,
    Contact_Name,
    Contact_Email,
    Contact_Phone,
    Sponsor_Tier
FROM Sponsor
WHERE Active_Flag = 1;
GO
```

Section 4I: View 8: All Tracks (1 Table, No WHERE, SELECT *)

Returns every column from the Track table with no filter applied.

```
USE JCF_Student15_DB;
GO
```

```
CREATE VIEW vw_AllTracks AS
SELECT *
FROM Track;
GO
```

Section 5: Lab 4 - Stored Procedures

Lab 4 was writing 5 stored procedures. Two of them take parameters, one rebuilds a summary report table for performance, one is a maintenance procedure that cleans up expired sponsor data, and one does a cascading delete. The drop statements at the top let the whole file run more than once without errors.

Section 5A: Drop Existing Stored Procedures

```
IF OBJECT_ID('sp_GetDriverRaceHistory', 'P') IS NOT NULL DROP PROCEDURE
sp_GetDriverRaceHistory;
IF OBJECT_ID('sp_UpdateCarStatus', 'P') IS NOT NULL DROP PROCEDURE sp_UpdateCarStatus;
IF OBJECT_ID('sp_RefreshRaceReport', 'P') IS NOT NULL DROP PROCEDURE
sp_RefreshRaceReport;
IF OBJECT_ID('sp_DeactivateExpiredSponsors', 'P') IS NOT NULL DROP PROCEDURE
sp_DeactivateExpiredSponsors;
IF OBJECT_ID('sp_DeleteDriverEntries', 'P') IS NOT NULL DROP PROCEDURE
sp_DeleteDriverEntries;
GO
```

Section 5B: Stored Procedure 1: sp_GetDriverRaceHistory (accepts parameter)

Takes a driver license number and returns all race history for that driver. You pass in the license number and get back every race they entered along with their result, points, and prize money.

```
USE JCF_Student15_DB;
GO

CREATE PROCEDURE sp_GetDriverRaceHistory
    @LicenseNumber VARCHAR(25)
AS
BEGIN
    SELECT
        D.Driver_FirstName,
        D.Driver_LastName,
        R.Race_Name,
        R.Race_Date,
        RE.Starting_Position,
        RE.Qualifying_Time,
        RR.Finish_Position,
        RR.Points_Earned,
        RR.Finish_Status,
        RR.Prize_Won
    FROM Driver D
    INNER JOIN RaceEntry RE ON D.Driver_ID = RE.Driver_ID
    INNER JOIN Race R ON RE.Race_ID = R.Race_ID
    LEFT JOIN RaceResult RR ON RE.RaceEntry_ID = RR.RaceEntry_ID
    WHERE D.License_Number = @LicenseNumber
    ORDER BY R.Race_Date;
END
GO
```

Section 5C: Stored Procedure 2: sp_UpdateCarStatus (accepts parameters)

Takes a car number and a new status code and updates that car's record. Instead of writing an UPDATE statement by hand every time a car changes status, just call this with the car number and new code.

```
USE JCF_Student15_DB;
GO

CREATE PROCEDURE sp_UpdateCarStatus
    @CarNumber VARCHAR(10),
    @NewStatus CHAR(2)
AS
BEGIN
    UPDATE Car
    SET CarStatus_Code = @NewStatus
    WHERE Car_Number = @CarNumber;
END
GO
```

Section 5D: Stored Procedure 3: sp_RefreshRaceReport (prefill report table)

Drops and rebuilds a summary report table called RaceReportSummary from Race, Track, RaceEntry, Driver, Car, and RaceResult. The idea is this runs once a day so queries against the report table are fast instead of running a big multi-table join every time.

```
USE JCF_Student15_DB;
GO

CREATE PROCEDURE sp_RefreshRaceReport
AS
BEGIN
    IF OBJECT_ID('RaceReportSummary', 'U') IS NOT NULL
        DROP TABLE RaceReportSummary;

    SELECT
        R.Race_Name,
        R.Season_Year,
        R.Race_Date,
        R.Weather_Summary,
        T.Track_Name,
        T.Country_Name,
        D.Driver_FirstName,
        D.Driver_LastName,
        C.Car_Number,
        RE.Starting_Position,
        RR.Finish_Position,
        RR.Points_Earned,
        RR.Finish_Status,
        RR.Prize_Won
    INTO RaceReportSummary
    FROM Race R
    INNER JOIN Track T ON R.Track_ID = T.Track_ID
    INNER JOIN RaceEntry RE ON R.Race_ID = RE.Race_ID
    INNER JOIN Driver D ON RE.Driver_ID = D.Driver_ID
    INNER JOIN Car C ON RE.Car_ID = C.Car_ID
    LEFT JOIN RaceResult RR ON RE.RaceEntry_ID = RR.RaceEntry_ID;
END
GO
```

Section 5E: Stored Procedure 4: sp_DeactivateExpiredSponsors (maintenance)

Checks the Join_Team_Sponsor table for contracts whose end date has already passed and sets those sponsors to inactive. This keeps the Sponsor data accurate without having to check and update manually.

```
USE JCF_Student15_DB;
GO

CREATE PROCEDURE sp_DeactivateExpiredSponsors
```

```

AS
BEGIN
    UPDATE Sponsor
    SET Active_Flag = 0
    WHERE Sponsor_ID IN (
        SELECT Sponsor_ID
        FROM Join_Team_Sponsor
        WHERE Contract_EndDate < GETDATE()
    )
    AND Active_Flag = 1;
END
GO

```

Section 5F: Stored Procedure 5: sp_DeleteDriverEntries (cascading delete, accepts parameter)

Takes a driver license number and deletes all race records for that driver. It has to delete from RaceResult before RaceEntry because of the foreign key — if you try to delete from RaceEntry first it throws an error since RaceResult still references it.

```

USE JCF_Student15_DB;
GO

CREATE PROCEDURE sp_DeleteDriverEntries
    @LicenseNumber VARCHAR(25)
AS
BEGIN
    DECLARE @DriverID UNIQUEIDENTIFIER;
    SET @DriverID = (SELECT Driver_ID FROM Driver WHERE License_Number =
@LicenseNumber);

    DELETE FROM RaceResult
    WHERE RaceEntry_ID IN (
        SELECT RaceEntry_ID FROM RaceEntry WHERE Driver_ID = @DriverID
    );

    DELETE FROM RaceEntry
    WHERE Driver_ID = @DriverID;
END
GO

```

Section 5G: EXEC Test File

The SQL below tests all 5 stored procedures to confirm they run without errors.

```

USE JCF_Student15_DB;
GO

```

```
-- Test SP 1: pull all race history for driver LIC-001 (Liam Carter)
EXEC sp_GetDriverRaceHistory @LicenseNumber = 'LIC-001';
GO

-- Test SP 2: update car number 9 from Under Repair back to Active
EXEC sp_UpdateCarStatus @CarNumber = '9', @NewStatus = 'AC';
GO

-- Test SP 3: rebuild the race report summary table
EXEC sp_RefreshRaceReport;
GO

-- Test SP 4: deactivate any sponsors whose contracts have expired
EXEC sp_DeactivateExpiredSponsors;
GO

-- Test SP 5: delete all race entries for driver LIC-020 (Tomas Novak)
EXEC sp_DeleteDriverEntries @LicenseNumber = 'LIC-020';
GO

PRINT 'All stored procedure tests executed.';
GO
```